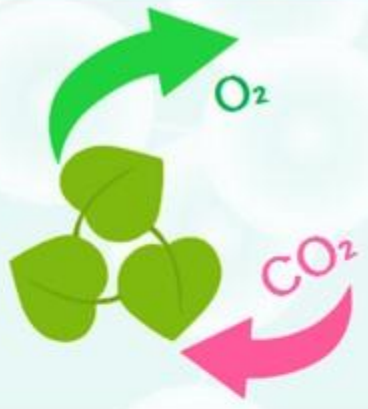




BIOLOGY



CLASSIFICATION OF ORGANISMS - 2

MODULE - 14



WHY DO WE CLASSIFY DIFFERENT FORMS OF LIFE?



- *Each form of life has different characteristics.*
- *Helps to understand Evolutionary changes between organisms*
- *Helps to identify and group newfound organisms under a proper category.*
- *Easy study of the characteristics.*



KINGDOM ANIMALIA

1 PORIFERA

2 CNIDARIA

3 PLATYHELMINTHES

4 NEMATODA

5 MOLLUSCA

6 ANNELIDA

7 ARTHROPODA

8 ECHINODERMATA

9 PROTOCHORDATA

10 VERTEBRATA

*Mnemonic - Prince Charles Plays New Music
And Always Eats Perfect Vegetables.*



1

PHYLUM PORIFERA



*Eg – Calcareous sponge
Demosponge*

- Non motile
- Pores all over the body
- Very minimal cell differentiation
- Exclusively marine



2

PHYLUM CNIDERIA



Eg - Jellyfish
Sea-anemone
Anthozoa

- These are diploblastic animals
- Exclusively marine
- Some species live in colonies like corals
- Some live solitary like Jellyfish and sea anemone

Diploblastic- body possess two types of cell layers, one inner endoderm and other outer ectoderm



3

PHYLUM PLATYHELMINTHES



Eg - tapeworm
liverfluke

- Three layers of cells- ectoderm, mesoderm and endoderm from outer to inner.
- These are triploblastic animals.
- They have bilateral symmetry
- No true internal body cavity or coelom in which well differentiated organs can be accommodated.
- These organisms are flat dorsiventrally
- Either free living or parasites.

Bilateral symmetry the body means that the left and the right halves of the body have the same design



4

PHYLUM NEMATODA



Eg - *Ascaris*
Enoplea

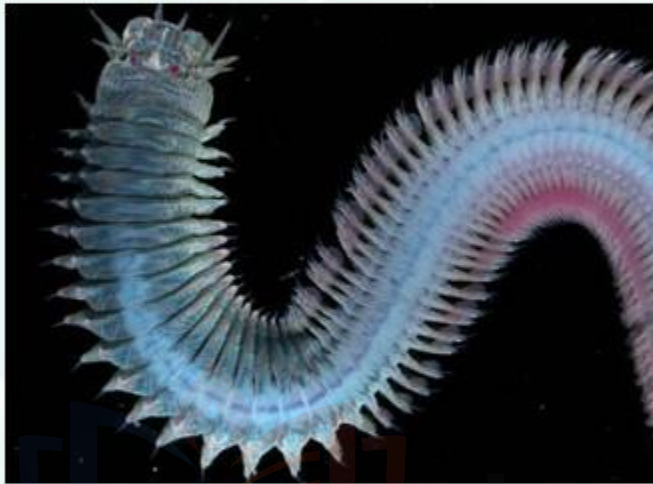
- Bilaterally symmetrical
- Triploblastic body
- Body is cylindrical rather than flat
- Presence of a pseudocoelom
- Mainly parasitic

- Filarial worms – Elephantiasis
- Intestinal worm - roundworm



5

PHYLUM ANNELIDA



Eg - Earthworm
Leech
Polychaete

- Bilaterally symmetrical
- True body cavity
- True organs are packaged in the body structure
- Extensive organ differentiation
- Body is segmented.
- Can be freshwater, marine water or land animals



6

PHYLUM ARTHROPODA



- Bilaterally symmetrical
- Body is segmented into head, Thorax and abdomen
- Jointed legs
- Open circulatory system.
- Have coelomic cavity

Eg - all insects
crustaceans
arachnids



7

PHYLUM MOLLUSCA



- Bilateral symmetry
- Coelomic cavity is reduced
- Open circulatory system
- Kidney like organs for excretion
- Foot for moving around

Eg -

Snail

Octopus

Squid

Cuttlefish

Learning Space



8

PHYLUM ECHINODERMATA



- Spiny skinned organisms
- Marine animals
- Five point radial symmetry
- Water driven tube system through their bodies

Eg -

Starfish

Sea urchin

Sea cucumber



9

PHYLUM PROTOCHORDATA



Eg - *Balanoglossus*
Herdmania

- Bilaterally symmetrical
- Notochord is seen at some stages of their life.
- Muscles for easy movement
- Marine animals

Notochord is a long rod like support structure that runs along the back of the animal separating the nervous tissue from the gut



10

PHYLUM VERTEBRATA



- Also known as Phylum Chordata
- Consists of all animals who have a spinal cord and vertebral column
- They have an anterior end (mouth) and a posterior end (anus)
- Presence of Central Nervous System with well developed organ system.
- Further divided into five main 'Class'



1

CLASS PISCES



Eg - Shark
Electric fish
Rohu, Catla

- Notochord is replaced by vertebrae
- Heart is two chambered.
- Endoskeleton can be cartilaginous or bony
- Exoskeletons are usually covered by scales
- Can be freshwater or marine

Chondrichthyes have skeletons made up of cartilage
Osteichthyes have skeletons made up of bones



2

CLASS AMPHIBIA



- Amphibians have a double life on land and water
- Both gills and lungs
- Three chambered heart
- External fertilization
- Coldblooded

Eg - Frog, Toad
Newt, Salamander
Caecilians





Eg -
Turtles
Chameleon
Crocodiles
Snakes

- Egg laying or oviparous
- Cold blooded
- Have tough dry scales in their body with some having colour changing ability
- Have three chambered heart except crocodile having four chambered heart.
- First animals adapted to live entirely on land.





- Warm blooded
- Spindle shaped body
- Forelimbs are modified for flying
- Four chambered heart
- Well developed nervous system
- External communication like cooing
- Prenatal care is well developed.



5

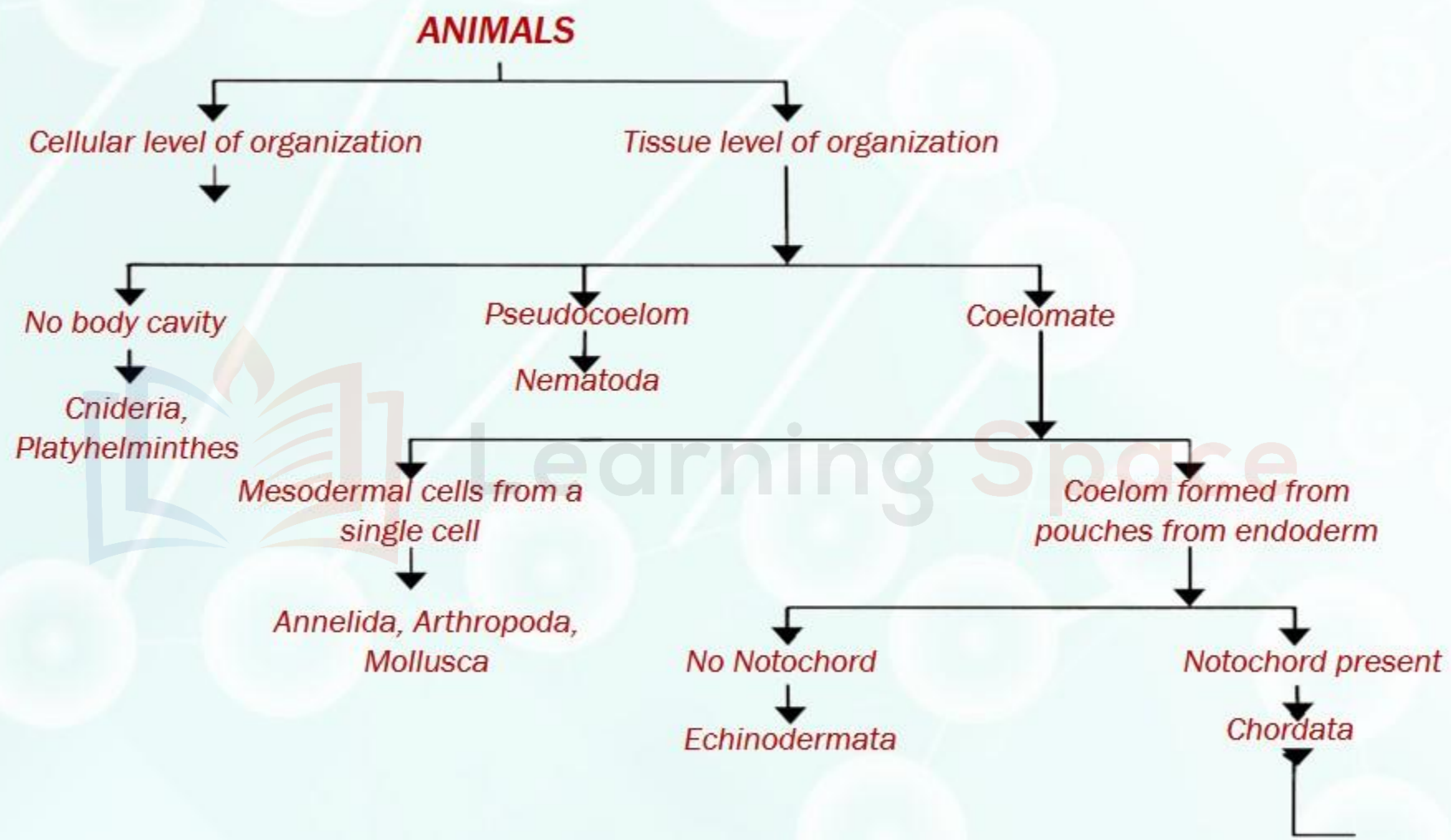
CLASS MAMMALIA

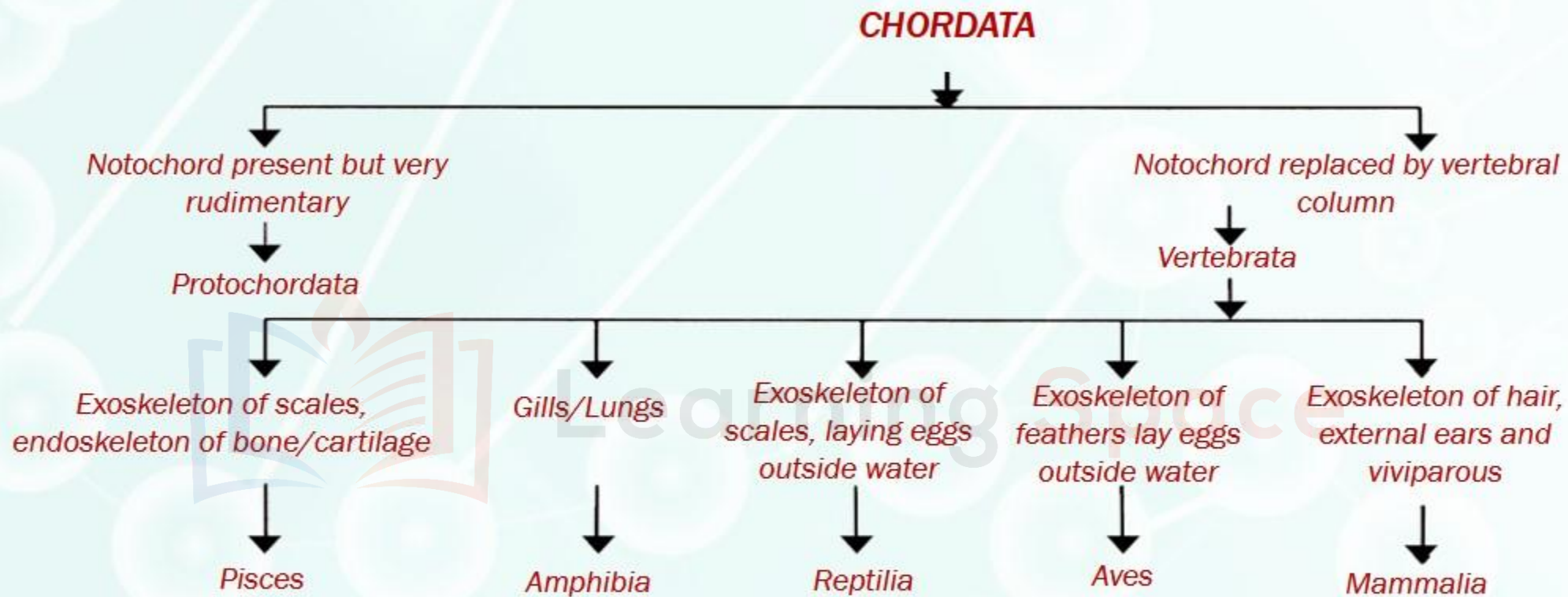


Eg -
Rodents
Carnivores
Primates

- Four chambered heart and lungs present
- Give birth to young ones or viviparous(except species of monotremes)
- Skin is covered with hair evolved from scales
- Most **mammals are placental** – the embryo develops internally in a uterus to the mother by a placenta.
- **Some are marsupial** – in the very early part of the embryonic development the young are born and completes its development in the mother's pouch.
- Only humans can sleep on their backs.







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